

Example

Me
4 September 2026

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Problem 1.1

Find all odd prime numbers p such that both $\frac{p+1}{2}$ and $\frac{p^2+1}{2}$ are perfect square numbers.

Insight

Two things here: first that a prime p as a factor is quite nice to work with, second that with $a \mid b$ always consider the magnitudes of both sides.

Solution

We have $p+1 = 2a^2$ and $p^2+1 = 2b^2$. Then $p(p-1) = 2(b-a)(b+a)$. There are two possibilities here. First, $p \mid b-a$. However this is impossible - from the two conditions given $p > a$ and $p > b$. So it can only be that $p \mid b+a$. Since $a+b < 2p$, $a+b = p$ and $p-1 = 2(b-a)$. Then it follows that $p+1 = 4a$. Since $p+1 = 2a^2$, we conclude that $(a, b) = (2, 5)$ is the only solution and $p = 7$.

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Problem 2.1

Determine whether there is an infinite sequence of prime numbers $p_1, p_2, p_3 \dots$ such that for any positive integer k , either $p_{k+1} = 2p_k + 1$ or $p_{k+1} = 2p_k - 1$.

Insight

There are two ways to go about doing this. I was at a loss as to how to start, but it's always helpful to do a bit of modular analysis to give you some preliminary information. So I took $p \pmod{3}$, and it turned out to be helpful as $p_i \pmod{3}$ then remains the same throughout the sequence, either -1 or 1 . What this allows you to do is express two possible sequences: $\{3l-1, 6l-1, 12l-1 \dots\}$ or $\{3m+1, 6m+1, 12m+1 \dots\}$. Most likely it then will not work. You can also realise the usual way to go about sequences does apply here, i.e. you can express it as $p_{k+1} + 1 = 2(p_k + 1)$. Then obtain $p_k = 2^{k-1}(p_1 + 1) - 1$. Since only the power of 2 changes here, you can apply FLT, where $2^{q-1} \equiv 1 \pmod{q}$ and thus $2^{q-1} \cdot 2^{k-1}(p_1 + 1) - 1 \equiv 2^{k-1}(p_1 + 1) - 1 \pmod{q}$. Actually this is the easier solution.

Solution

The idea here is to take a term p_i and have it divisible by some q . Then since there are convenient powers of 2 we can invoke FLT on, we will force a relation to a larger term p_j , such that it is equivalent to $p_i \pmod{q}$, i.e. it is divisible by q . This works since it is larger than p_i , meaning it is composite.

Solution left as an exercise to the reader.

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Problem 3.1

Let S be a set of $6n$ points along a straight line, among which $4n$ points chosen randomly are coloured red and the other $2n$ points coloured blue. Show that one may always find a line segment containing exactly $3n$ points in S , among which $2n$ are red and n blue.

Insight

This is essentially this question: Show that there are 1000 consecutive positive integers containing exactly 10 prime numbers. Use the sliding window technique.

4 Geom

Having obsessively pored over olympiad geom (and left out A/C/N), there are a few tricks I observed to be widely employed. Many of these techniques are really just configurations that have gained popularity (in what seems to be US contests). Some of the questions will employ many techniques, so they will be repeated in the list below. The list is not exhaustive; it merely serves as an example to the technique. But of course I recommend you try them out first without the technique in mind to see if you can derive it on your own.

4.1 Techniques

Technique 4.1.1 Radical axis

Use length relations for power of a point, then identify if on radical axis.

e.g: Iran TST 2011/1, IMO 2009/2, USA TSTST 2017/1, USA TST 2011/4, **GOTEEM 2**, CGMO 2015/6

Technique 4.1.2 Identifying orthocentres

As the name suggests^a.

e.g: Sharygin 2013, ELMO 2018/4, ELMO SL 2019 G3

^aI don't believe this is a technique in and of itself, for some it will come naturally but at least for me I need to single it out and remind myself. Also orthocentres appear very often for some reason.

Technique 4.1.3 Duality of excentres and orthocentres

4.2 Configs

Config 4.2.1 Nine-point circle

Whenever you spot a *midpoint and a foot of altitude*, or the *midpoint of OH* is brought up, then you should consider the nine-point circle. This will often be tied in with Technique 4.1.1, so you then have to notice the many concyclities, including the nine-point circle.

e.g: USAMO 2015/2, USA TSTST 2017/1